

Justify all answers. Submit as a single PDF.

1. (10 points) Suppose that we roll a pair of fair dice until the sum of the numbers on the dice is seven. What is the expected number of times we roll the dice?
2. (10 points) Let A be an event. Define $\mathbf{1}_A$, the **indicator random variable** of A , by

$$\mathbf{1}_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Show that the expected value of the indicator random variable equals the probability of the event, that is, $\mathbb{E}[\mathbf{1}_A] = \Pr(A)$.

3. When m balls are distributed into n bins uniformly at random:

- (a) (10 points) What is the probability that the first bin remains empty?
- (b) (10 points) What is the expected number of balls that fall into the first bin?
- (c) (10 points) What is the variance of the number of balls that fall into the first bin?
- (d) (10 points) What is the expected number of bins that remain empty?

Hint: Use indicator random variables. For bin 1, let $I_i = \mathbf{1}_{\{\text{ball } i \text{ is in bin 1}\}}$ and set $X = \sum_{i=1}^m I_i$; use linearity of expectation for parts (b)–(c). For empty bins, let $J_j = \mathbf{1}_{\{\text{bin } j \text{ is empty}\}}$ and set $Y = \sum_{j=1}^n J_j$ to compute $\mathbb{E}[Y]$ in part (d).

4. The law of total expectation states that, if the sample space S is the disjoint union of events $S_1, S_2, \dots, S_n$ and X is a random variable, then

$$\mathbb{E}[X] = \sum_{j=1}^n \mathbb{E}[X \mid S_j] \Pr(S_j).$$

- (a) (10 points) Prove the law of total expectation.
- (b) (10 points) Use the law of total expectation to find the average weight of a breeding elephant seal, given that 12% of the breeding elephant seals are male and the rest are female, and the expected weights of a breeding elephant seal are 4200 pounds for a male and 1100 pounds for a female.
5. (10 points) Let X be a random variable on a sample space S such that $X(s) \geq 0$ for all $s \in S$. Show that

$$\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

for every positive real number a . This inequality is called **Markov's inequality**. *Hint: Use the indicator $\mathbf{1}_{\{X \geq a\}}$ and the inequality $\mathbf{1}_{\{X \geq a\}} \leq X/a$. Take expectations.*

6. Suppose that the number of cans of soda pop filled in a day at a bottling plant is a random variable with an expected value of 10,000 and a variance of 1000.
- (a) (10 points) Use Markov's inequality (previous exercise) to obtain an upper bound on the probability that the plant will fill more than 11,000 cans on a particular day.
 - (b) (10 points) Use Chebyshev's inequality to obtain a lower bound on the probability that the plant will fill between 9000 and 11,000 cans on a particular day.
7. A golf tournament offers a \$1,000,000 prize if a single amateur makes a hole-in-one on a par-3 (one swing). Use the underwriting probability

$$p = \Pr(\text{hole-in-one}) = \frac{1}{12,500} = 0.00008 \quad (0.008\%).$$

The insurer charges a premium $P = \$180$ and pays \$1,000,000 if the shot goes in (else pays nothing).

- (a) (10 points) Compute the insurer's expected profit per contest.
- (b) (10 points) Compute the Kelly-optimal fraction for the insurer.